# Histology of the Ovary and Organs of the Ovum Pathway

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Block: Endocrinology/Urology /Reproduction
Thread: Anatomy

Conflict of Interest: Dr. Macnow has no Conflicts of Interest to disclose.

## Textbook Links

Histology: A Text and Atlas, 9th ed.–Chapter 23: Female Reproductive System

## Learning Objectives

By exam time, you should be able to do the following:

1. Discuss the structural and functional anatomy of the ovary and organs involved in ovum release or implantation, including how they change over different life stages and the hormonal regulation of these changes.
2. Describe and identify the cells and tissues of the ovary and organs involved in ovum release or implantation by histology.
3. Discuss the process of oogenesis from primordial follicle through mature oocyte, including the associated morphologic changes.
4. Compare and contrast how the ovary and organs involved with ovum release or implantation function as, and are regulated by, endocrine glands; and identify the impact of hormonal change via microscopy.

## STUDY QUESTIONS

1. Compare and contrast the histological appearance of the primordial and primary follicle.

2. Why might ectocervical and vaginal epithelial cells need glycogen stores?

3. Kartagener syndrome is a ciliopathy characterized by a mutation in dynein, rendering cilia immotile. What effect might this have in the uterus and/or adnexa? Which organ(s) are likely to be affected?

## A NOTE ON TERMINOLOGY

You may see differences in how various resources (and our own lecturers!) discuss the stages of follicular development. This probably represents a difference between sources for the material. For this anatomy lecture, I am using the assigned textbook (Pawlina & Ross, *Histology: A Text and Atlas, chapter 23*) as my source. In this reference, it provides a disclaimer:

> "*Histologically, three basic types of* ***ovarian follicles*** *can be identified on the basis of developmental state:*
>
> ***Primordial follicles***
> ***Growing follicles****, which are further subcategorized as primary and secondary (or antral) follicles*
> ***Mature follicle*** *or* ***Graafian follicles***
>
> *Some histologists and clinicians identify additional stages in the continuum of follicular development.*"

The Pawlina text defines a primary follicle as having simple cuboidal cells (early) which subsequently stratify (late); a secondary follicle as containing a fluid-filled antrum. However, other sources may use slightly different criteria for determining these different levels of maturation.

For our purposes, we're earmarking this as a possible point of confusion for students and will make sure to avoid assessment items that might lead to having more than one correct answer based on the diversity of sources.

## A NOTE ON SUPPLEMENTARY RESOURCES

Starting in academic year 2025–2026, this learning activity was changed from a histology lab to a lecture. This was done in response to student requests for more structure regarding this content. However, the resources from the session as a histology lab still exist!

The materials are included on Canvas as optional, supplementary material that give you an opportunity to practice the content within this session.

*Note: Additional information, in a more "conversational" or "narrative" format, are provided with the PDF version of your presentation slides.*

## A. THE OVARY

1. The **ovary** is the structure producing **ova (oocytes, eggs)**. It is divided into the **cortex**, where the oocytes develop in **follicles**, and the **medulla**, with fibrovascular tissue supporting the cortex and providing it blood supply and venous drainage.
2. The **ovary** is covered with a layer of **mesothelium** called the **mesovarium**, corresponding to the portion of the **peritoneum** that covers the organ. The peritoneum covering the internal reproductive structures is called the **broad ligament**; the mesovarium is a subportion of the broad ligament. The mesovarium is a simple squamous to cuboidal layer of epithelium.
3. The ovary has a capsule of dense irregular connective tissue called the **tunica albuginea**.
4. Follicles contain oocytes and develop through several stages:
    a. **Primordial follicles** contain a primary oocyte (arrested in prophase of meiosis I) surrounded by a single layer of squamous follicular cells. All follicles are in this stage at the time of birth, until **menarche** (first menstrual cycle, in puberty).
    b. Approximately every 28 days, a group of about 20–30 primordial follicles are recruited to become primary follicles. **Primary follicles** are surrounded by cuboidal cells, which proliferate under the stimulation of **follicle-stimulating hormone (FSH)**.
        i. Early primary follicles are **unilaminar**, with one layer of follicular cells.
        ii. Late primary follicles are **multilaminar**, with several layers of follicular cells (which are renamed to be called **granulosa cells**).
        iii. The **zona pellucida**, a layer of glassy-appearing glycoprotein, forms between the oocyte and the granulosa cells.
    c. **Secondary follicles** contain an **antrum**, filled with **follicular fluid (liquor folliculi)**.
        i. The background stroma of the ovarian cortex is the **theca folliculi**; it becomes more organized into layers around secondary follicles.
        ii. The **theca interna** surrounds the granulosa cells and is separated from them by a distinct basal lamina. It contains cells with ample, pale-staining cytoplasm (because they are producing steroid hormones). They are stimulated by **luteinizing hormone (LH)** to produce **androstenedione**, which is subsequently converted into

**estrogens** by the enzyme **aromatase** in granulosa cells.

iii. The **theca externa** surrounds the theca interna and has spindle-shaped cells that are involved in contracting and mechanically extruding the oocyte during ovulation.

d. The **mature (Graafian) follicle** has a large fluid-filled antrum and is over 10 mm in size. It continues to have the theca interna and externa, as described before. The granulosa cells organize into specific groups and the oocyte continues to mature:

i. The layers of granulosa cells immediately surrounding the oocyte correspond to the **corona radiata**.

ii. The granulosa cells surrounding the antrum correspond to the **zona granulosa**.

iii. A few anchoring groups of cells, the **cumulus oophorus**, connect the oocyte and its corona radiata to the zona granulosa cells on the wall of the follicle.

iv. The oocyte completes meiosis I and arrests in **metaphase II** right before ovulation. During telophase and cytokinesis I, a **polar body** will be produced, containing a haploid complement of chromosomes but minimal cytoplasm.

5. By day 14 of the menstrual cycle, generally just one (occasionally two, very rarely more) mature follicle will be selected to undergo ovulation.

a. Follicles that are not chosen for ovulation undergo **atresia** (degeneration). The granulosa cells will become pyknotic and dissolve, and the oocyte will lose its nuclear features. A layer of thickened basement membrane, the **glassy membrane**, forms between the degenerated granulosa cells and the theca interna.

b. The follicle(s) that are chosen for ovulation undergoes characteristic changes in response to a surge of **luteinizing hormone (LH)** at day 14 of the menstrual cycle.

i. As the follicle breaches the tunica albuginea and mesovarium, a transparent swelling called a **stigma (macula pellucida)** is formed.

ii. The stigma ruptures, releasing the follicular fluid. The oocyte is released, covered with the corona radiata and some associated cumulus oophorus.

iii. The oocyte and its associated granulosa cells and follicular fluid are guided into the **oviduct** by its **fimbriae**. Ciliated cells line the oviduct (discussed more in later sections), which create a current to draw the oocyte into the **infundibulum** of the oviduct (**fallopian tube** or **uterine tube**).

6. Once ovulation has occurred, only a subset of the granulosa cells and the theca cells remain in the ovary, where they develop into the **corpus luteum**.
    a. The granulosa cells mature into **granulosa lutein cells** and increase in size as they ramp up production of **progesterone**, **estrogens**, and **inhibin A**.
    b. The theca interna cells mature into **theca lutein cells**, which produce androgen precursors used by the granulosa lutein cells to synthesize their hormones, as well as producing some progesterone themselves.
    c. If fertilization occurs, the blastocyst will produce **human chorionic gonadotropin (*β*-hCG)**, which is a luteinizing hormone analogue (that is, it will stimulate the same receptor, keeping the corpus luteum alive). In essence, the blastocyst triggers the corpus luteum to produce the progesterone it requires to survive in the endometrial cavity.
    d. If fertilization does not occur, the corpus luteum will involute to form a **corpus albicans** after about ten days; essentially, a small nodule of fibrotic tissue marking where the follicle and its corpus luteum had previously existed.

## B. THE UTERINE (FALLOPIAN) TUBE (OR OVIDUCT)

1. The uterine tube can be subdivided into several sections:
    a. The **infundibulum** is a widened portion at the proximal end of the tube, containing finger-like projections (**fimbriae**) at its open end.
    b. The **ampulla** is the longest portion of the uterine tube and is the usual site of fertilization, if it occurs.
    c. It narrows down to a thin **isthmus** and meets the uterus at the **uterotubal junction**.
    d. There is an intramural segment that runs through the smooth muscle wall of the uterus to reach the endometrial cavity.
2. Like other luminal organs, the uterine tube has a mucosa, containing an epithelium overlying a lamina propria, layers of muscle, and an outer coat of serosa:
    a. The tubal epithelium is simple columnar though may take on a crowded, even pseudostratified look. It contains two major cell types:
        i. **Ciliated columnar cells**, which sweep in the direction of the uterine cavity to establish a current through the tube and guide the passage of the ovum. Ciliated cells are most numerous at the infundibular/ampullary end of the tube.
        ii. **Non-ciliated columnar (peg) cells** produce nutrient rich secretions that protect the oocyte and help activate spermatozoa. They are

most numerous at the isthmus/intramural end of the tube.

b. There are two layers of smooth muscle: an inner circular and outer longitudinal layer.

c. The **serosa** contains neurovascular structures and is covered with mesothelium (the same portion of peritoneum referred to as the broad ligament). The portion surrounding the tube is the **mesosalpinx**.

## C. THE UTERUS

1. The uterus can be divided into three tissue layers:
   a. The **mesometrium** is the **serosa**; the portion of the broad ligament that covers the uterus. It is the same mesothelium described before.
   b. The **myometrium** contains woven fascicles of smooth muscle and constitutes the majority of the organ.
   c. The **endometrium** contains the glandular tissue lining the inside of the uterus.
2. The endometrial cavity is the open space in the center of the uterus. It is continuous with the **endocervical canal** running through the uterine cervix.
   a. The **internal (cervical) os** is the narrow opening between the endometrial and endocervical spaces.
   b. The **external (cervical) os** is the narrow opening between the endocervix and the vaginal canal.
3. The myometrium can be organized into layers:
   a. There is an inner and outer layer of smooth muscle running roughly parallel to the axis of the organ.
   b. There is a layer of spirally-oriented smooth muscle between these other two layers, the **stratum vasculare**, that contains vascular plexi.
   c. During pregnancy, the myometrium becomes hypertrophic and hyperplastic and there is increased deposition of collagen. The vasculature also becomes more prominent.
4. The endometrium can be divided into two layers:
   a. The **stratum basalis** is the more basally located layer and remains in place and relatively unchanged throughout the menstrual cycle.
      i. It contains small rings of glandular cells in a simple columnar epithelium, embedded in somewhat cellular lamina propria (stroma).
      ii. It is the stem cell compartment for the overlying stratum functionalis.

b. The **stratum functionalis** is the more apically located layer and undergoes significant changes throughout the menstrual cycle; it is ultimately sloughed off during menses.

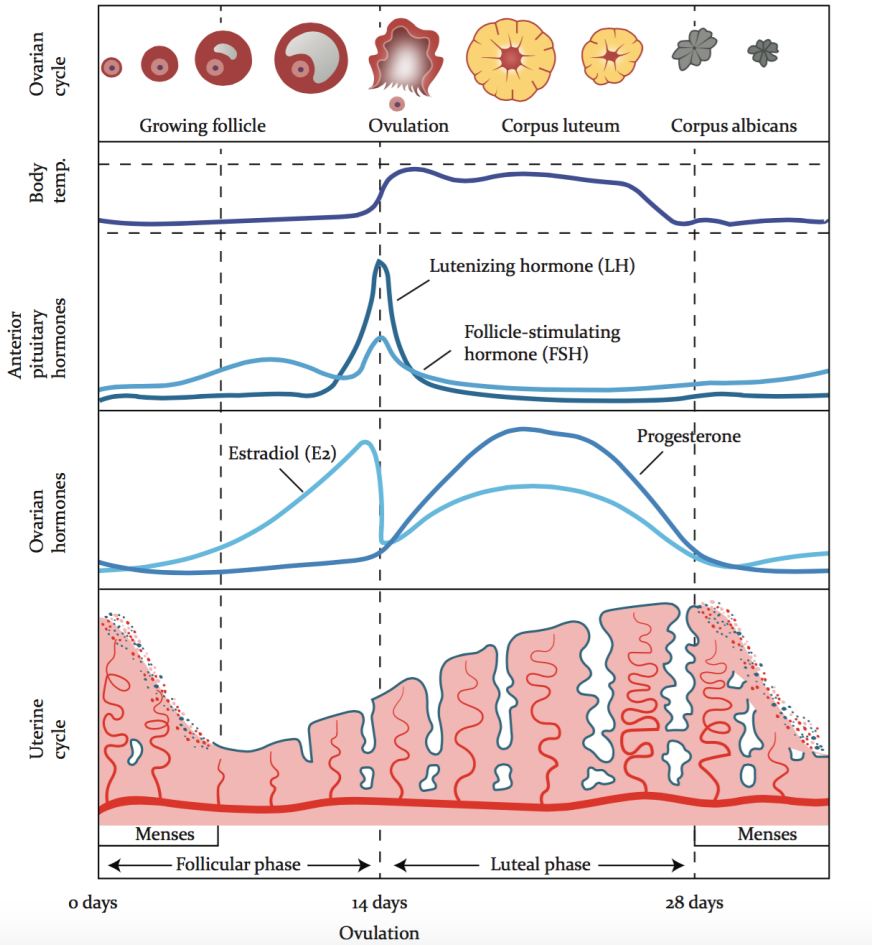
   i. During **proliferative phase** (approximately days 6–14 of the menstrual cycle), it contains small rings or tubules of simple columnar epithelium embedded in relatively cellular lamina propria, like the stratum basalis. Estrogen drives the proliferation of these glands, and the development of **spiral arteries** that provide blood supply to this tissue.

   ii. During **secretory phase** (approximately days 14–28 of the menstrual cycle), there are several changes to the stratum functionalis predominantly in response to progesterone. The glands become more complex and corkscrew-shaped; their cytoplasm gets lighter in color due to the presence of glycogen-rich mucus. The spiral arteries become more coiled. The lamina propria becomes more edematous in some areas and more dense and pinker in others (**predecidual change**).

   iii. During **menses** (approximately days 0–5 of the menstrual cycle), progesterone drops and this withdrawal leads to deterioration of the stratum functionalis. The stroma condenses into dense, basophilic balls of tissue. Neutrophils infiltrate into the tissue and hemorrhage ensues with breakdown of the spiral arteries; the stratum functionalis is sloughed off.

   iv. If fertilization and implantation occur, the endometrium will develop into **decidua**: the predecidual change seen in the secretory phase will give way to large cells with glassy, eosinophilic cytoplasm and prominent cell borders, due to glycogen storage and hormone production.

5. Let's line things up between the anterior pituitary, ovary, endometrium, and the classic menstrual cycle diagram shown below:




a. Days 5–14: ovarian follicular phase/endometrial proliferative phase
- Anterior pituitary produces follicle-stimulating hormone, which stimulates ovarian follicles to develop (follicular phase).
- Granulosa cells of the developing follicle produce estrogens (from androstenedione from the theca interna cells); estrogen stimulates endometrial thickening and spiral artery formation.
- The stratum functionalis of the endometrium is in proliferative phase.

b. Day 14: ovulation
- Anterior pituitary produces luteinizing hormone surge, which stimulates ovulation within the ovary
- Oocyte and associated granulosa cells (corona radiata and cumulus oophorus) are extruded through tunica albuginea and mesovarium
- The stratum functionalis of the endometrium is completing proliferative phase

c. Days 14–28: ovarian luteal phase/endometrial secretory phase
- Anterior pituitary hormones suppressed by high levels of ovarian hormones

- Oocyte traverses oviduct; granulosa cells left behind and theca cells become luteinized, forming the corpus luteum, which produces progesterone; progesterone stimulates secretory phase of the endometrium.
- The stratum functionalis of the endometrium is in secretory phase, with corkscrew-like glands, prominent and tortuous vessels, areas of stromal edema, and areas of stromal condensation (predecidual change)

d. Days 0–5: menses
- Anterior pituitary hormone levels begin to rise again, with a drop in ovarian hormone suppression
- Oocyte carried out with the sloughing endometrium; the corpus luteum begins involution, eventually turning into a corpus albicans.
- The stratum functionalis of the endometrium breaks down, with stromal condensation, hemorrhage, and neutrophilic infiltrate; and is sloughed off as menstrual material

## D. THE CERVIX AND VAGINA

1. The **uterine cervix** is a cylindrically-shaped portion of the uterus and has specific regions:
   a. The **endocervix (endocervical canal)** is lined by simple columnar, mucus-producing epithelium. It also contains large, cyst-like spaces filled with mucus.
   b. The **ectocervix** faces the vaginal canal and is lined by nonkeratinizing stratified squamous epithelium.
   c. The **transformation zone** is situated between the endo- and ectocervices and has metaplastic squamous epithelium. It is the region most vulnerable to **human papillomavirus (HPV)** infection.
2. The underlying stroma of the uterine cervix is composed predominantly of dense, fibrous connective tissue with a minor contribution of smooth muscle. It is rich in elastic fibers, enabling cervical effacement (thinning) and dilation (opening) during labor and delivery.
3. The **vagina** is lined with nonkeratinizing, but glycogenated, stratified squamous epithelium.
   a. The glycogen imparts quite a bit of white space to the cytoplasm of the cells and is a source of nutrition for the normal vaginal flora, like *Lactobacillus* spp.
   b. The vaginal lamina propria does not have its own mucous glands; its mucus derives from secretions from the endocervix.

## Answers to Study Questions:

***These answers may not be all encompassing; rather they may be examples of the types of thought processes needed for application of the material.***

1. Compare and contrast the histological appearance of the primordial and primary follicle.

   *The primordial follicle has a large primary oocyte arrested in prophase I and a ring of squamous follicular cells around the oocyte. They are in the cortex, surrounded by stromal cells and are waiting to restart the maturation process. The primary follicle has resumed the maturation process. The oocyte may be noticeably larger and begins to advance through meiosis I. The follicular cells are now cuboidal; near the end of the primary follicle stage, they may be stratified and are now called granulosa cells. The oocyte is forming the zona pellucida in a primary follicle and the stratified granulosa cells are divided from the surrounding theca (stromal) cells by a defined basement membrane. In addition, the theca interna cells take on morphology consistent with their role as steroid hormone producers.*

2. Why might ectocervical and vaginal epithelial cells need glycogen stores?

   *The microenvironment of the lower reproductive tract (that is, vagina and cervix), must be kept in check for several reasons. One is to guard against infection and the other is to protect sperm that enter the vaginal canal. Remember, this reproductive system is open to both the external environment (via the cervix) and the peritoneal cavity (via the uterine tube). As cervical cells slough off and die, the glycogen is used by commensal bacteria—especially* Lactobacillus *spp.—to generate lactic acid. This keeps the pH low and prevents colonization of the lower reproductive tract by pathogens.*

3. Kartagener syndrome is a ciliopathy characterized by a mutation in dynein, rendering cilia immotile. What effect might this have in the uterus and/or adnexa? Which organ(s) are likely to be affected?

   *Kartagener syndrome, or primary ciliary dyskinesia, will render the ciliated cells of the uterine tube immotile. This will slow or stop the transit of the oocyte from the ovary into the uterine tube, as well as down the length of the tube toward the endometrium. This may cause infertility (about 61% of individuals with uteri and adnexa who have primary ciliary dyskinesia have infertility). Additionally, if fertilization does occur, the lack of ciliary motility also increases the risk of ectopic pregnancy (implantation outside of the endometrial cavity, usually within the uterine tube).*